

Technology Stack

Date	08 June 2026
Team ID	
Project Name	EcoTrack - Personal Carbon AI Footprint Tracker
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and

1. Frontend Presentation Layer:
 - Framework: React.js (Vite Build Tool)
 - Styling: Vanilla CSS & Tailwind CSS (Dark Mode / Glassmorphism)
 - Charts & Icons: Chart.js, React-Gauge-Chart, Lucide Icons
2. Backend Application Layer:
 - Web Server: Flask Python Web Server (Python 3.10+)
 - Modules: REST API Controllers, Cors Policy Handling, Dotenv Config
3. Database & Authentication Layer:
 - Database: Firebase Firestore Cloud Database
 - Authentication: Firebase Auth Service & User Profile Storage
4. AI & Third-Party Integrations:
 - AI LLM API: Groq LLM API (llama-3.3-70b-versatile)
 - Verification Gateway: Carbon Interface API for emissions datasets

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	<i>[Specify technology]</i>	<i>[Explain why this was chosen (e.g., fast processing, existing team expertise)]</i>
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	<i>[Specify technology]</i>	<i>[Explain why this was chosen (e.g., handles unstructured data, strong relational mapping)]</i>
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	<i>[Specify technology]</i>	<i>[Explain why this was chosen (e.g., easy scalability, cost-effective hosting)]</i>
5	Version Control & CI/CD	<i>[Specify technology]</i>	<i>[Explain why this was chosen (e.g., seamless collaboration, automated testing)]</i>

	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	<i>[Specify technology]</i>	<i>[Explain the specific feature this enables in your project]</i>